# The final visual cortex model (erjpu): every stage, its channels, the features it emits and where the nonlinearities are

No learned weights: every filter is set analytically or from a fixed seed, and ridge regression is the only fitted step. Feature counts are the model's own output layout (36 blocks, 9,996 features); nonlinearity badges follow its channel definitions and the description in the paper appendix.

## Input

RGB image
(LAION or NSD photo)

**Pixel compression** (P)

intensity raised to a power

**Opponent channels**

luminance L, red−green RG, blue−yellow BY (linear mix)

opponent channels

## V1-like stage

30 channels at 3 octaves (image pyramid: o0 fine, o1, o2)

**16** **luminance Gabor energies** (Q)

8 orientations (0–157.5°) × 2 scales

**2** **luminance blobs** (R)

bright and dark, difference of Gaussians

**8** **color Gabor energies** (Q)

RG and BY × 4 orientations

**4** **color blobs** (R)

red, green, yellow, blue

**Then each channel is divided** (D)

**by a pool of local energy**

(intermediate maps also keep small and large blob sizes, 36 maps in all)

## Output and ridge regression

Each block is normalized separately, then all 36 blocks are concatenated into one vector per image:

**9,996 features**

By computational family (each feature traced to its source channels):

| | |
|---|---|
| Edges | 3,562 |
| Texture statistics | 2,280 |
| Color | 2,148 |
| Corners and contours | 935 |
| Luminance curvature | 670 |
| Mixed | 243 |
| Global layout | 158 |

### Ridge regression

one linear map from the 9,996 features to each voxel, fit on each subject's unique images

### Order of operations

compression → energy or rectification → normalization → products, tuning → pooling → linear ridge regression

### Nonlinearity badges

- (P) power-law compression
- (Q) quadrature energy √(even² + odd²)
- (R) half-wave rectification max(±x, 0)
- (D) divisive normalization x / (σ + pool)
- (M) multiplication of channel pairs
- (T) analytic tuning curves of the shape index
- (S) pooling: mean, spread (std), peak (max)

Grids are foveated: cell size grows with eccentricity. "Spread" is the spatial standard deviation within a cell.

## Modules: intermediate maps

### V1 maps pooled on foveated grids (S)

*from: V1-like stage*

30 channels per grid cell; cells are finer near the centre of gaze.

### Raw color layout (S)

*from: opponent channels*

L, RG and BY averaged on a 6×6 grid, with no filtering.

### Texture boundaries (filter–rectify–filter) (Q) (S)

*from: V1 energy maps, one octave coarser*

A second Gabor bank applied to the energy maps and squared, so it responds where patterns of oriented contrast change.

### Orientation field and contour curvature (M) (R) (S)

*from: V1 energies at 3 octaves*

13 channels: 2nd and 4th orientation harmonics and their strength (junctions), curvature split by sign into +x, −x, +y, −y, |curvature|, iso- and cross-orientation surround.

### Edge phase (R) (S)

*from: even and odd Gabor responses at 3 octaves*

9 channels: bright line, dark crease, line vs edge, phase coherence, fine vs coarse energy, brightness gradient split into +x, −x, +y, −y.

### Surface geometry (T) (R) (S)

*from: smoothed luminance derivatives (Hessian, structure tensor) at 5 octaves*

14 channels: 5 shading shapes from the shape index (cup, rut, saddle, ridge, dome), corner and edge strength, edge coherence, isophote curvature split by sign, |Lxx−Lyy|, |2Lxy|, red−green and blue−yellow 2D structure.

### Co-occurrences (texture statistics) (M) (S)

*from: normalized channels of the stages above*

Products of channel pairs averaged over space, in the spirit of Portilla–Simoncelli texture statistics.

### Global layout (R) (S)

*from: summary channels and a 32-pixel image*

Coarse paths of contrast, light and dark extrema that persist across thresholds, left–right symmetry and centre versus periphery.

## Pooled output features

| | |
|---|---|
| o0: mean on 9×9 grid | 2,430 |
| o0: spread and peak on 5×5 | 750 |
| o1: mean on 6×6 | 1,080 |
| o1: spread and peak on 3×3 | 270 |
| o2: mean on 4×4 | 480 |
| o2: spread and peak on 2×2 | 120 |
| whole image: mean and spread | 60 |
| module total | 5,190 |

| | |
|---|---|
| 6×6 grid × 3 channels | 108 |
| module total | 108 |

| | |
|---|---|
| 3×3 grid | 720 |
| whole image: mean and spread | 160 |
| module total | 880 |

| | |
|---|---|
| o0 and o1: 4×4 grids | 416 |
| o2: 2×2 grid, mean and spread | 78 |
| module total | 494 |

| | |
|---|---|
| o0 and o1: 4×4 grids | 288 |
| o2: 2×2 grid, mean and spread | 54 |
| module total | 342 |

| | |
|---|---|
| o0: mean 4×4 + spread 3×3 | 350 |
| o1: mean 4×4 + spread 3×3 | 350 |
| o2: mean 3×3 + spread | 210 |
| o3: 2×2 grid, mean and spread | 84 |
| o4: mean and spread | 28 |
| module total | 1,022 |

| | |
|---|---|
| V1 channels within each scale | 536 |
| V1 channels within image quadrants | 264 |
| curvature, phase, geometry, color | 312 |
| persistence from octave to octave | 192 |
| fine channels × their coarse parents | 192 |
| geometry channels across fine scales | 144 |
| geometry channels across coarse scales | 114 |
| module total | 1,754 |

| | |
|---|---|
| coarse contrast paths | 80 |
| persistent light and dark extrema | 2 |
| symmetry, centre vs periphery | 54 |
| symmetry of geometry channels | 70 |
| module total | 206 |